KREISEL

BATTERY PACK

SOFTWARE GUIDE

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Legal

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Please read this first!

Thank you for choosing a high-tech product from KREISEL Electric!

The Kreisel Battery Pack (referred to as the battery in the following) is controlled by a Battery Management System (BMS).



For intended and safe use follow the instructions in this document.

Keep this document as a reference for later use.

Before configuring and connecting the BMS to the vehicle control unit (VCU), read and observe the information in this document and and the product specific documentation of the battery, in particular the safety instructions.

This will give you a comprehensive introduction to the special features of the system in order for you and others to identify and avoid potential hazards.



This Software Guide contains important additional information to the Technical Documentation of the Kreisel Battery Pack and is only valid in conjunction with it.

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List of abbreviations

Abbr.	Meaning
BMS	Battery Management System
CAN	Controller Area Network
CSC	Cell Sensor Circuit
DBC	data base CAN
DC	Direct Current
Dept.	Department
HV	High Voltage
MC	Main controller
PMB	Power measurement board
PP	Power Prediction
SC	Safety controller
SOC	State of Charge
SOH	State of Health
T	Terminal
VCU	Vehicle Control Unit

1 General

1.1 Purpose of the document

This document describes the communication interface of the BMS and gives instructions to the basic operating principles of the battery system. It is an essential part of the product and must be available to commissioning and service personnel at all times.



Failure to follow the instructions in this document may result in fire, personal injury and/or property damage.

Persons configuring and connecting the BMS to the vehicle control unit (VCU) must read, understand, and follow the instructions in this document and observe the safety and warning notices.

1.2 Symbols and notes

The following symbols and instructions warn of possible injury to persons or damage to property or provide you with work aids.



Warning sign

This warning sign is used to alert you to potential risks of injury. Observe all safety and warning instructions following this symbol to avoid possible injury – including fatality.

The signal words **DANGER**, **WARNING** and **CAUTION** are used for different danger levels and indicate the severity of the danger.



DANGER!

Denotes an immediately hazardous situation that will lead to serious injury – including fatality – if not avoided.



WARNING!

Denotes a potentially hazardous situation that can cause severe injury – including fatality – if not avoided.



CAUTION!

Denotes a potentially hazardous situation that can result in minor to moderate injury if not avoided.

The signal word *BEWARE* is used to indicate an imminent risk of damage to property.

BEWARE
Denotes a potentially threatening situation that can result in damage to property if not avoided.



A note contains useful information for efficient and trouble-free operation.



A link points to applicable documents.

1.3 Formatting and characters

The formatting and characters used in this document have the following meanings:

- Identifies a bullet list (1st level)
  - Identifies a bullet list (2nd level)

1. Identifies a step (1st level)
   Follow the sequence of steps!
   a. Identifies a step (2nd level)
      Follow the sequence of steps!
   › Describes a result.



The decimal separator is a comma and the thousand separator a point according to ISO 80000-1.

1.4 Applicable documents

- CAN Matrix
  - “KE_BMS_Customer_v1.dbc”
  - “KE_BMS_Customer_v1.xls”

1.5 Technical Support

For questions about installation, configuration and operation of the battery, or in case of malfunction, please contact our Technical Support. We are happy to answer your questions and look forward to your feedback. You can reach our Technical Support during our business hours.

TECHNICAL SUPPORT

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support@kreiselelectric.com

2 Basic Principle of Operation

The BMS ensures the safe operation of the lithium-ion cells of the battery and continuously monitors the cell parameters. Based on the acquired information, the BMS informs the higher-level vehicle control unit (VCU) of the permissible charge and discharge currents. These values and the state of charge (SOC) are essential for safe operation of the battery.

If any emergency error occurs, the BMS ensures the integrity of the battery and puts it into a safe state by opening the main contactors.

All communication between the BMS and the VCU takes place via CAN.

For further information please refer to the CAN Interface Description in section 5.

2.1 Wakeup Options

The battery can be configured with two different wakeup options:

- T15 wakeup
- CAN wakeup

This configuration is hardware dependent and cannot be changed after delivery.

Refer to the specific technical documentation for the wakeup-option of the Kreisel Battery Pack.

2.1.1 T15 Wakeup

Figure 1 shows the state transitions and sequences of the BMS with the corresponding CAN signals for the T15 wakeup.

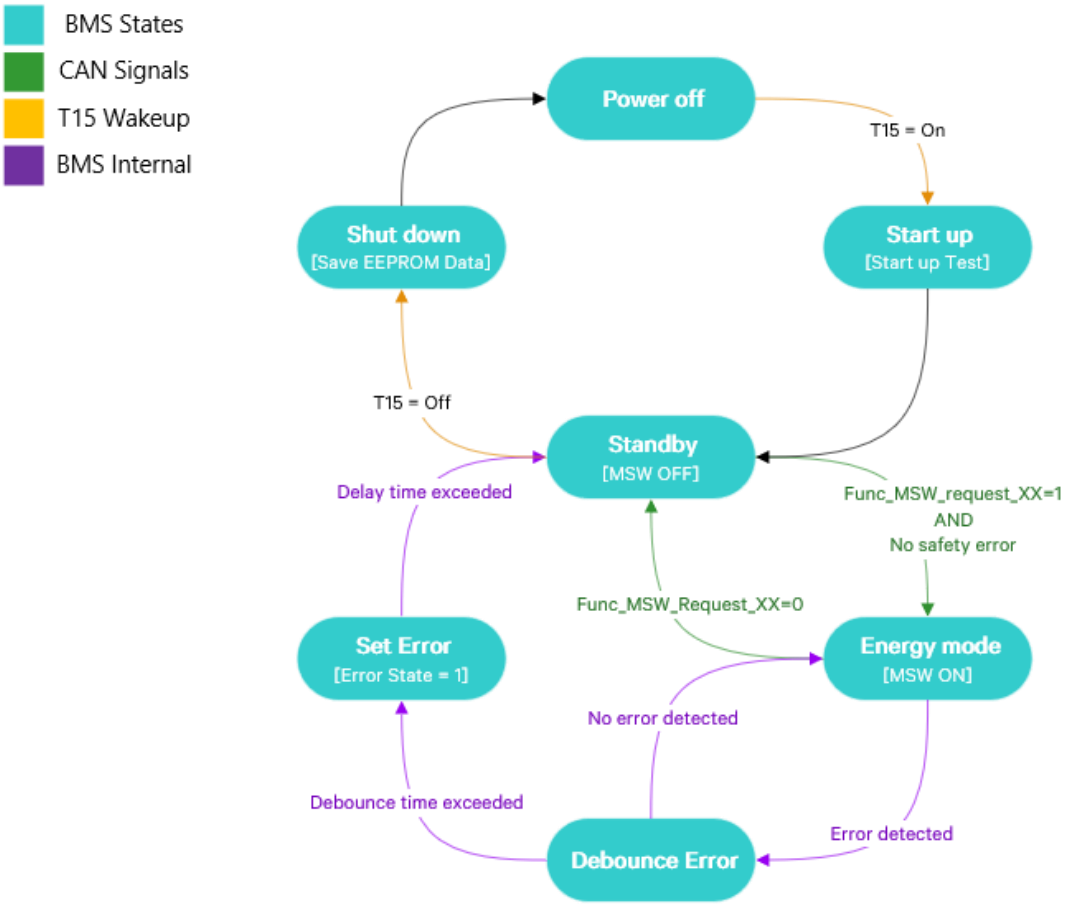


Figure 1 - BMS State diagram wakeup-option T15 with corresponding CAN signals

2.1.2 CAN Wakeup

Figure 2 shows the state transitions and sequences of the BMS with the corresponding CAN signals for the CAN wakeup.

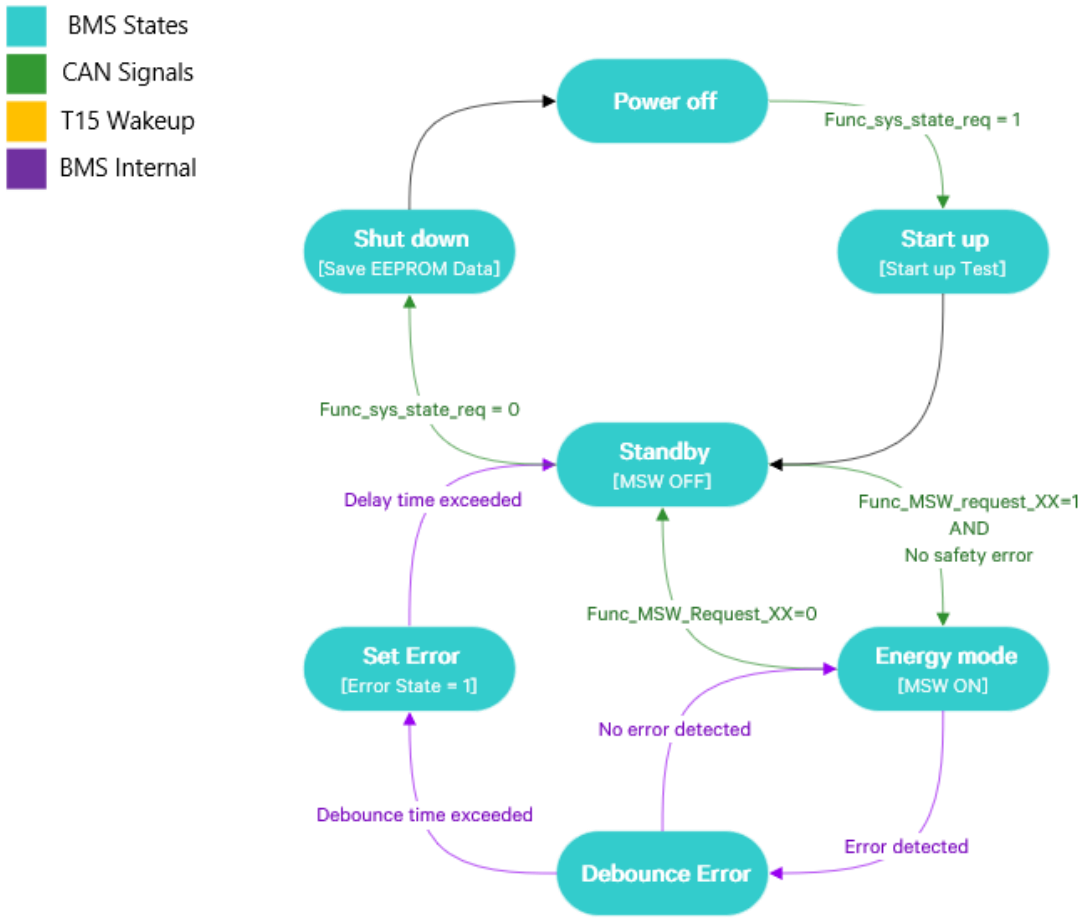


Figure 2 - BMS State diagram wakeup-option CAN with corresponding CAN signals

2.2 BMS Start up

1. The following conditions must be met:
   a. The BMS must be powered by the low voltage network of the vehicle via T30 and T30c.
2. Wake up the BMS via
   - T15 for T15 wakeup
   - CAN signal “Func_sys_state_req”=1 for CAN wakeup (100ms cyclic)
3. The BMS sends the CAN signals
   - “BMS_start_up_ready”=1,
   - “BMS_stay_alive”=1,
   - “BMS_wakeup_active”=1

   when the start-up process is complete.
4. The BMS is in standby mode.

2.3 BMS Shut down

⚠ WARNING!

Functional hazard by switching off the KL30!

If the BMS is not switched off as described, important BMS data cannot be saved and are lost. In this case, the BMS-CAN data are not displayed correctly after a restart.

- Switch off the BMS in a controlled manner (see below).

BEWARE

Switching off under load reduces the service life of the main contactors.

- Make sure that the battery current is zero (main contactors open) before shutting down the battery.

1. Switch off wakeup signal
   - T15 for T15 wakeup
   - CAN signal “Func_sys_state_req”=0 for CAN wakeup
     Last single shot of message EXT_FuncRequest (7FFh)
     “Func_sys_state_req”=0, afterwards stop sending the message

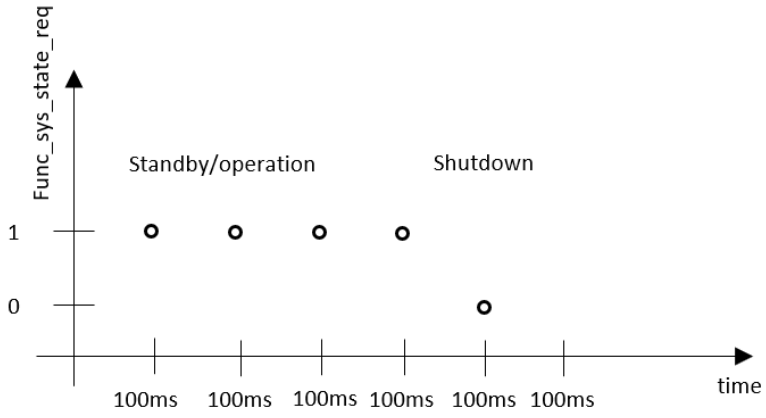


Figure 3 – BMS Startup/Shutdown with Func_sys_state_req

2. The BMS opens the main contactors if not already open.
3. The BMS shuts down in line with its internal protocols.
4. CAN communication from the BMS ends.
   - › The BMS is in power off mode (sleep mode).

Make sure that no other components send any further CAN messages to the BMS. Otherwise the BMS cannot enter sleep mode.

With T15 wakeup it is possible to keep the BMS in standby or energy mode with the CAN signal “Func_sys_state_req”=1. The BMS will shut down if the Signal is set to 0.

2.4 Energy Mode

1. The BMS must be in standby mode (refer to section 2.2)
2. The VCU uses the CAN “Func_MSW_Request_XX” signal to send a request to the BMS to close the main contactors.

   Options in a multi battery system:

> ⚠ **WARNING!**
>
> **Compensation currents may damage the batteries.**
>
> - The voltage difference between each battery must be less than 3V if multiple batteries are connected.
> - Make sure the difference is less than 3V before trying to close all contactors of the multiple battery system.

   - The BMS handles the main switch close request automatically. To do so, the CAN “Func_MSW_Request_XX” signals (where XX indicates the specific battery) must be set to 1 for all the batteries, which should close their contactors. The BMS closes the main switches of the battery with the lowest voltage first.
   - It is also possible that the VCU decides the close order of the pack main switches. For this, the CAN signals “Func_MSW_Request_XX” must be set in the preferred order. It is recommended, to send the close request for the next pack after the close response was received.
3. If no constraints are present, the BMS starts the pre-charge process. If pre-charging is successful, the BMS closes the main switches.
4. The BMS returns the main switch state in the CAN “BMS_XX_MSW_closed” signal.

   › The BMS is in energy mode.

> *BEWARE*
>
> **Switching off under load reduces the service life of the main contactors.**
>
> - Avoid opening the main contactors during normal operation.

5. If the BMS does not receive “Func_MSW_Request_XX”=1 every 100ms or the VCU sends “Func_MSW_Request_XX”=0, the BMS opens the main contactors, regardless of the current running through the battery at the moment.

2.5 Error Mode

1. The BMS detects a critical error and the defined debounce time is exceeded.



If the error is no longer detected, during debounce time, the BMS remains in energy mode.

2. The BMS sends the CAN error signal “BMS_EmergencyState_XX”=1.
3. After a defined delay, the BMS opens the main contactors regardless of the current running through the battery at the moment.

   › The BMS is in standby mode

The BMS sends the system events (errors, warnings) in the CAN message mBMS_ROE. The message is not sent cyclically. A message is sent once for each error/warning that occurs. The user/VCU needs to focus on events coming from the Main Controller and Safety Controller components (EVT_Component: 1=MC; 2=SC). For further details refer to section 5.2.

2.5.1 Reset Error Mode

It is possible for the VCU to reset BMS error mode by sending the CAN signal “Func_reset=1”. Reset will be done, if the main switches are opened, otherwise a reset request will be denied. (“BMS_reset_denied=1”)

3 BMS Functions

The basic functions of the BMS are listed in this section.

3.1 Power Prediction

The BMS has several current limits based on the cell chemistry used and the actual operating conditions. These limits are defined by Kreisel Electric and are not user configurable. They can be found in the battery technical documentation.

> ⚠ **WARNING!**
>
> **Risk of fire and poisoning!**
>
> Violating the defined limits may cause damage to the battery, and may result in a hazardous incident.
>
> - The VCU must ensure that all components on the HV network operate within the power prediction limits.

The BMS sends the actual permissible current limits for

- Continuous Discharge [A]
- Continuous Charge [A]

via the CAN message mBMS_PP to the VCU.

The BMS cannot send the values for possible peak current. If it is necessary to use currents above continuous current, it is to be calculated as noted below:

Ipeak: average current value during peak current usage
Irest: average current value during resting phase
Icont(T): maximum allowed continuous current depending on cell temperature
Ipeak(T): maximum allowed peak current depending on cell temperature
tpeak: peak duration
trest: necessary resting duration (at Irest) before the next peak current usage

$$Irest = \frac{[Icont(T)(tpeak + trest)] - (Ipeak\ tpeak)}{trest}$$

$$trest = -\frac{\left(Ipeak - Icont(T)\right) tpeak}{Irest - Icont(T)}$$

These calculations have to respect following limitations:

- $0\ sec < tpeak \leq 10\ sec$
- $0\ \mathrm{sec} < trest$
- $Icont(T) < Ipeak \leq Ipeak(T)$
- $0\ A \leq Irest < Icont(T)$

The calculations are valid for charge and discharge current, but they are not independent. Meaning, that after a charge peak, a discharge peak is not allowed. After each peak, the resting period has to follow before another peak is allowed, no matter if it is charge or discharge.

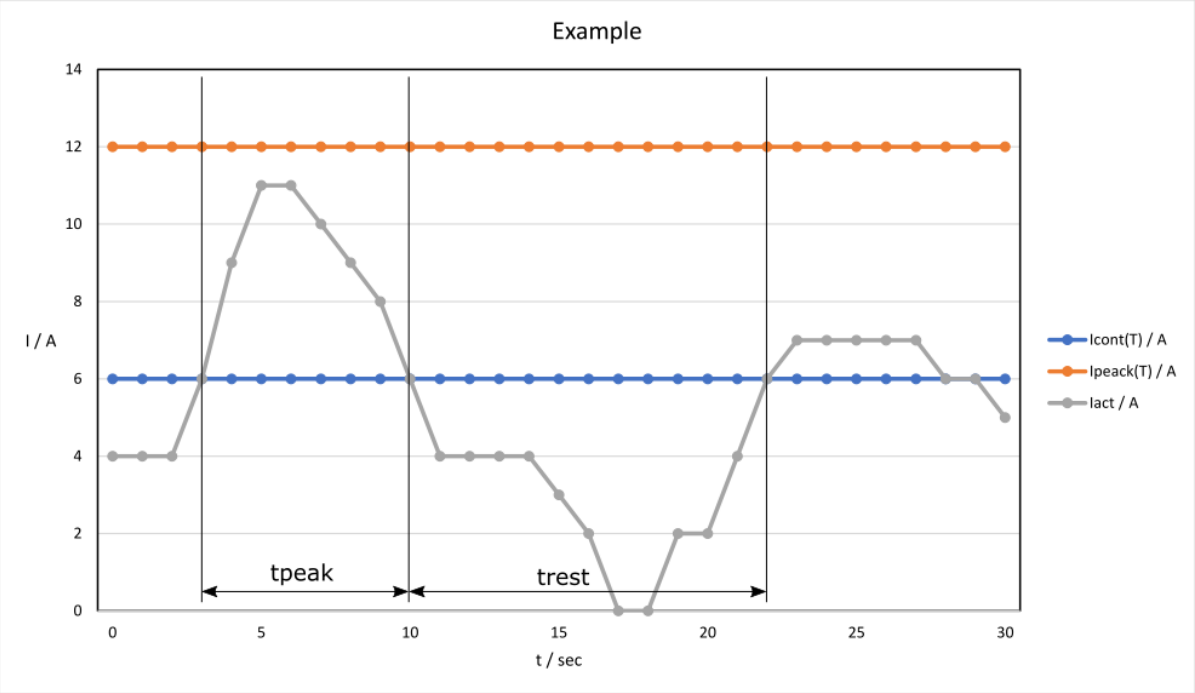


Figure 4 – Power Prediction example

3.2 SOH MSW

The BMS monitors the main contactor switching cycles and considers the conditions under which they take place. Based on this information, the BMS calculates the SOH of the main contactors. Open the main switches under high power will cause a high counter value as opening the main switches under current 0A. The BMS prevents the contactors from switching on again when the CAN signal "BMS_SOH_MSW_XX" reaches 0%.

3.3 Pre-charge

The battery includes a pre-charge circuit, which allows the BMS to charge the DC link capacity to avoid high inrush currents when switching on the main contactors.

The BMS always automatically performs a pre-charge procedure before closing the main switches. To verify that the capacitors are appropriately pre-charged, the BMS monitors the difference between battery voltage and traction voltage. During pre-charge the BMS monitors the pre-charge time and energy in order to avoid overheating of the pre-charge resistor.

BEWARE

The actual pre-charge time depends on the DC link capacity of the customer traction network.

BEWARE

Compensation currents may damage the batteries.

- The voltage difference between each battery must be less than 3V if multiple batteries are connected.

3.4 Isolation Monitoring

Isolation monitoring supervises the isolation resistance of the high voltage circuit of the vehicle.

The BMS uses its own isolation measuring device, which should be activated using the “func_iso_measurement” CAN signal.
The resistance value is measured before the main switches between HV+ and GND of the system and HV- and GND of the system. At opened main switches, the internal system is measured.
The BMS measures the resistance value of the complete system (including the internal measurement resistors) in kOhm.

⚠ **WARNING!**

Risk of electric shock!

Multiple isolation measuring devices can cause a measurement malfunction.

- In case of an external isolation measuring device, the BMS's internal isolation measuring must not be activated

In a multiple battery system, the isolation measurement will be handled by the BMS automatically. One of the batteries which has closed the main switches, will activate the isolation measurement, if an activate-request is available. At opened main switches, the minimal value of all packs will transmitted on the ESS-CAN.

- During an active isolation measurement, the BMS sends the CAN signal “BMS_iso_meas_act”=1.
- Over the CAN signal “BMS_iso_value_valid”, the BMS inform the VCU if the resistance value is valid or not.
- The isolation resistance is available in the CAN signal “BMS_Iso_Resist” in kΩ. Only use the value, if the “BMS_iso_value_valid” signal is true. A short debounce time is recommended.

While the activation signal is being sent, the BMS continuously measures the isolation resistance.

The BMS outputs a warning (eSC_STAT_ISO_RES_WARNING) and an error (eSC_ERR_ISO_RESISTANCE). The warning and error thresholds are listed in the specific technical documentation of the Kreisel Battery Pack.

⚠ WARNING!

Risk of electric shock!

Isolation faults can lead to the housing of the battery carrying a high voltage.

- Make sure that the VCU initiates a safe shutdown of the battery system.

3.5 SOC Recalibration

Normally SOC recalibration starts automatically.

The CAN signal “BMS_SOC_RecalibState_XX” indicates the SOC recalibration state and is 0 at start-up time. After the first recalibration, the state changes to 1.

Main condition that the BMS starts SOC recalibration automatically is that the actual battery current is < +/- 200mA.
Optionally it can be started manually with the CAN signal “Func_SOC_recalib”, not restricted to the actual battery current.

3.6 Balancing

Balancing compensates for the varying self-discharge rates that the cells may develop during their service lives. The BMS balancing functionality helps to align the cell voltage of each individual module with the minimal module voltage value.

The need for balancing is application dependent and the balancing procedure shall only be initiated, if:

- the battery system is in a quiescent state with no current flowing,
- all cell voltages are at a stabilized level and
- the SOC has reached a narrow pre-determined band.

3.6.1 Starting Balancing

Conditions for starting balancing:

- The BMS must always be in standby or energy mode
- The minimal cell voltage must be higher than 3800mV.
- The desired threshold voltage in the CAN signal “Threshold_voltage” must be higher than 3800mV and should be close to the minimal cell voltage.

1. To start balancing, the CAN signal “Func_balancing” must be set to 1 and the CAN signal “Threshold_voltage” must be set to a valid value.

   › If balancing starts successfully, the BMS sends the “BMS_balancing_act”=1 response signal.

   › Balancing stops automatically after 720 min or if all cells are discharged to the “Threshold_voltage”
   It can always be stopped manually by setting the the CAN signal “Func_balancing” back to 0.

The need for balancing depends on the application and it is therefore the customer's responsibility to start balancing. It is recommended to start balancing after every charging process to hold the cell voltage difference as low as possible. To achieve the best performance of the battery,the maximum cell voltage difference should be < 10mV.

BEWARE

During balancing, the external power supply must be monitored to prevent a deep discharge of the low voltage battery (12V/24V).

3.6.2 Balancing Example

For Example, Balancing can be started after charging of the Kreisel Battery Pack. A possible strategy would be:

1. Charging the battery until the battery is full and a quiescent current is reached (e.g. < +/- 100mA per cell, consider number of parallel connected cells according to battery technical documentation)
2. Starting Balancing if required (U_Cell_,max – U_Cell_min) > 10mV @Battery full
   Threshold_voltage = U_Cell_min + 5mV
   (all Cells that are above this threshold voltage, will be discharge through balancing resistors)
3. Stop balancing when cell voltage difference is <5mV or battery power is required

3.7 Multiple Battery System

⚠ WARNING!

Risk of fire!

There is a risk of fire, if a single battery in a multiple battery system fails and the system continues with the full current.

- Make sure that the VCU initiates a safe shutdown of the multiple battery system.

The BMS is capable of controlling eight batteries in a multiple battery system. To enable this functionality, each battery must be configured in accordance with the specific technical documentation of the Kreisel Battery Pack.

Controlling a multiple battery system is similar to controlling a single battery system. The main difference is controlling the main switches as described in section 0 and the error signals as described in section 5.2.

Furthermore, the battery leader of the multiple battery system summarizes all necessary battery values in the CAN message “mBMS_Bat_Info” and “mBMS_Cell_Info”. If the battery specific parameters are needed, one can use the corresponding battery CAN messages “mBMS_Bat_Info_XX” and “mBMS_Cell_Info_XX”, where XX indicates the specific battery (XX=Master; Slave1; … Slave7).

In case of only one battery, all master values (XX = Master) are valid.

4 Battery Safety

The battery system has been developed with a strong focus on functional safety.

BEWARE

The safe state of the battery system is defined as OFF with both main contactors and the pre-charge relay opened, separating the traction net from the high voltage.

The BMS switches to a safe state, if at least one of the following events occurs:

- Overvoltage of battery cells
- Undervoltage of battery cells
- Overtemperature of battery cells
- Battery overcurrent
- Loss of interlock signal
- Crash signal activated T30c
- Voltage supply outside of specified range
- Internal routine and start-up diagnostics fail

The defined limits depend on the battery and are preconfigured in accordance with the specific technical documentation of the Kreisel Battery Pack. These limits are defined by Kreisel Electric and are not user configurable.

⚠ WARNING!

Risk of fire and poisoning!

Violating the defined limits may cause damage to the battery, and may result in a hazardous incident.

- The VCU must ensure that all components on the HV network operate within the power prediction limits.

5 CAN Interface Description

The signals described in this document are formatted in line with the Intel Protocol.

Baud rate: 500kbit/s

All messages, except “mBMS_ROE”, are sent cyclically. The “mBMS_ROE” signal is transmitted only once when an error occurs. To send the error again, it must be triggered with “Func_send_roe_msg_again”.

The following CAN IDs are already used by the BMS:

101h, 102h, 110h, 111h, 112h, 113h, 114h, 115h, 116h, 117h, 118h, 130h, 131h, 160h, 161h, 1B0h, 1B1h, 200h, 201h, 250h, 251h, 2A0h, 2A1h, 2F0h, 2F1h, 340h, 341h, 390h, 391h, 700h, 701h, 7FCh, 7FEh, 7FFh.

In the following lists and the provided CAN database file only the relevant messages are listed.

5.1 mBMS_FuncStat (101h)

Message Name	Description	Value Range	Additional Info
BMS_start_up_ready	Start-up ready state	0: Start-up not done; 1: Start-up was done	
BMS_stay_alive	Self-holding state of battery	0: inactive; 1: active	
BMS_iso_meas_act	State of isolation measurement	0: inactive; 1: active	
BMS_balancing_act	Actual balancing state	0: inactive; 1: active	
BMS_reset_denied			
BMS_wakeup_active	Signs if rtc wake-up time is set	0: inactive; 1: active	
BMS_iso_value_valid	State of isolation measurement	0: inactive; 1: active	
BMS_MC_eep_write_finished	MC eeprom write finished flag	0: inactive; 1: active	
BMS_XX_available	Availability of battery XX	0: inactive; 1: active	
BMS_XX_MSW_Closed	Main contactor state of battery XX	0 = MSW open 1 = MSW closed	
BMS_EmergencyState_XX	Emergency state of battery XX	0: inactive; 1: active	
BMS_KL15_state	KL15 state	0: inactive; 1: active	
BMS_SOC_RecalibState_XX	SOC recalibration state of battery XX	0: inactive; 1: active	
Func_Output_No	BMS output number	[0; 15]	Currently not used

Message Name	Description	Value Range	Additional Info
Func_Output_Pack_No	BMS Pack number of the controlled output	[0; 2]	Currently not used
Func_Output_PWM_value	BMS output PWM value	[0; 100]	Currently not used

5.2 mBMS_ROE (102h)

Message Name	Description	Possible Values	Additional Info
EVT_Component	Event setting component		Multiplexor
EVT_Pack	Event battery pack information	[0; 7]	
EVT_ErrNr_CSC_XX_Master	Event error number CSC XX	[0x0; 0x3F]	XX ... [1; 16]
EVT_ErrNr_MC_Master	Event error number MC	[0; 7]	
EVT_ErrNr_PMB_Master	Event error number PMB	[0x0; 0x3F]	
EVT_ErrNr_SC_Master	Event error number SC	[0x0; 0x62]	
EVT_Context	Event Context. Further information on event.	[0x0; 0xFFFFFFFF]	
EVT_Categorie_Master	Error category	[0; 2]	
EVT_Timer_Master	Elapsed time since power up until event occurrence	[0; 1073741823]	

5.3 mBMS_Bat_Info (110h)

Message Name	Description	Value Range	Unit	Additional Info
BMS_I_Batt	Actual battery current	[-3276,8; 3276,7]	A	Negative values: Charge | Positive Values: Discharge
BMS_U_Batt	Overall battery voltage	[-3276,8; 3276,7]	V	Measured on battery side of switches
BMS_SOH_Cell	State of health of cells	[0;255]	%	Calculated on basis of cell resistance
BMS_SOH_MSW	State of health of main switch	[0;255]	%	Calculated on basis of switch counts (relating to actual current)
BMS_SOC	State of charge of battery	[0;100]	%	Calculated state of charge

5.4 mBMS_Bat_Info_XX (XX)

Message Name	Description	Value Range	Unit	Additional Info
BMS_I_XX	Actual battery current of battery XX	[-3276,8; 3276,7]	A	Negative values: Charge | Positive Values: Discharge

Message Name	Description	Value Range	Unit	Additional Info
BMS_U_XX	Overall battery voltage of battery XX	[-3276,8; 3276,7]	V	Measured on battery side of switches
BMS_SOH_CELL_XX	State of health of cells of battery XX	[0;255]	%	Calculated on basis of cell resistance
BMS_SOH_MSW_XX	State of health of main switch of battery XX	[0;255]	%	Calculated on basis of switch counts (relating to actual current)
BMS_SOC_XX	State of charge of battery of battery XX	[0;255]	%	

5.5 mBMS_Cell_Info (111h)

Message Name	Description	Value Range	Unit	Additional Info
U_Cell_avg	Average cell voltage based on the whole battery	[0; 65535]	mV	Voltage measured at every module
U_Cell_max	Maximum cell voltage based on the whole battery	[0; 65535]	mV	Voltage measured at every module
U_Cell_min	Minimum cell voltage based on the whole battery	[0; 65535]	mV	Voltage measured at every module
Cell_T_max	Maximum temperature of cell based on the whole battery	[-128;127]	°C	Temperature measured at least on 3 modules per Stack
Cell_T_min	Minimum temperature of cell based on the whole battery	[-128;127]	°C	Temperature measured at least on 3 modules per Stack

5.6 mBMS_CEll_Info_XX (XX)

Message Name	Description	Value Range	Unit	Additional Info
U_Cell_avg_XX	Average cell voltage based of battery XX	[0; 65535]	mV	Voltage measured at every module
U_Cell_max_XX	Maximum cell voltage based of battery XX	[0; 65535]	mV	Voltage measured at every module
U_Cell_min_XX	Minimum cell voltage based of battery XX	[0; 65535]	mV	Voltage measured at every module
Cell_T_max_XX	Maximum temperature of cell based of battery XX	[-128;127]	°C	Temperature measured at least on 3 modules per Stack
Cell_T_min_XX	Minimum temperature of cell based of battery XX	[-128;127]	°C	Temperature measured at least on 3 modules per Stack

5.7 mBMS_Trac_Info (114h)

Message Name	Description	Value Range	Unit	Additional Info
BMS_U_Trac	Battery voltage of the whole battery at closed main switches	[-3276,8; 3276,7]	V	Measured on system side of main switches
BMS_Iso_Resist	Actual isolation resistance value	[0;0]	k	Minimum Value of all batteries in a multiple battery system
BMS_PACK_No_R_Iso	Battery Pack number of shown ISO resistance value	[0-100]		

5.8 mBMS_PP (112h)

Message Name	Description	Value Range	Unit	Additional Info
PP_I_chrg	Actual possible charge current	[0; 6533.5]	A	Continuous
PP_Pwr_chrg	Actual possible charge power	[0;65535]	kW	
PP_I_dschrg	Actual possible discharge current	[0; 6553.5]	A	Continuous
PP_Pwr_dschrg	Actual possible discharge power	[0;65535]	kW	

5.9 mBMS_Temp_Info (115h)

Message Name	Description	Value Range	Unit	Additional Info
BMS_ext_Temp_TS1_max	Max. Temperature value Ext. Temp. Sensor TS1	[-40; 125]	°C	Differs from min only in multiple battery systems
BMS_ext_Temp_TS1_min	Min. Temperature value Ext. Temp. Sensor TS1	[-40; 125]	°C	
BMS_ext_Temp_TS2_max	Max. Temperature value Ext. Temp. Sensor TS2	[-40; 125]	°C	Differs from min only in multiple battery systems
BMS_ext_Temp_TS2_min	Min. Temperature value Ext. Temp. Sensor TS2	[-40; 125]	°C	

5.10 EXT_FuncRequest (7FFh)

Message Name	Description	Possible Values	Additional Info
Func_MSW_Request_XX	Open or Close request to main switches of battery XX	1: Close; 0: Open	
Func_iso_measurement	Request for isolation measurement	1: On; 0: Off	
Func_sys_state_req	System state request for BMS	0: do shutdown 1: stay alive	Not to be used with T15 (KL15)
Func_SOC_recalib	Recalibrate state of charge value request	0: inactive; 1: active	
Func_reset	Error reset request	0: no request; 1: reset request active	
Func_send_roe_msg_again	Request error message resend	0: no request; 1: request active	

5.11 5.11 EXT_BalancingRequest (7Feh)

Message Name	Description	Value Range	Unit	Additional Info
Func_balancing	Balancing activation request	0: inactive; 1: active		
Threshold_voltage	Threshold voltage	[0; 65535]	mV	Balancing voltage, must be higher than 3800mV

6 Error List

The mBMS_ROE message contains a multiplexer signal named “EVT_COMPONENT”.

“EVT_CATEGORIE_MASTER” differentiates between Safety Error (0x02) and Warning (0x00).

EVT_PACK designates the Battery Pack in which the error occurs.

6.1 SC Error Description

EVT_COMPONENT Value: 0x02

Error	ErrNr	Description (Short)
eSC_ERR_ACC_EMERGENCY	0x05	Not used
eSC_ERR_ACC_EMERG_TEST	0x39	Not used
eSC_ERR_ADC	0x3B	Error in SC ADC measurement.
eSC_ERR_BCC_COM	0x1D	Timeout of MC function request message detected.
eSC_ERR_BSC_OP_10D	0x55	SC continuous operation time more than 10 days.
eSC_ERR_BSC_OP_9D23H	0x54	SC continuous operation time more than 9 days and 23 hours.
eSC_ERR_CELL_OV_WARN	0x5B	Cell overvoltage warning limit reached
eSC_ERR_CELL_UV_WARN	0x5C	Cell undervoltage warning limit reached
eSC_ERR_CPU_TEST	0x14	Error occurred in SC CPU test.
eSC_ERR_CRASH	0x38	T30c (KL30c) value below low value.
eSC_ERR_CSC_DS_INTEROP	0x3D	CSC data set error detected
eSC_ERR_CSC_EMERG_SHORT_SUP	0x1C	Short circuit of CSC emergency line to supply detected.
eSC_ERR_CSC_EMERG_TEST	0x57	CSC emergency line test failed.
eSC_ERR_CSC_EMERGENCY	0x07	Low CSC emergency line detected.
eSC_ERR_CSC_INTERNAL	0x2C	CSC internal error; for details see sensor CAN.
eSC_ERR_CSC_INTEROP	0x33	Invalid CSC application version detected.
eSC_ERR_CSC_LIMITS	0x30	Defined limit violated at CSC; for details see sensor CAN.
eSC_ERR_CSC_CELL_OVERTEMPERATURE	0x2E	Cell overtemperature error limit reached.
eSC_ERR_CSC_OVERVOLTAGE	0x00	Cell overvoltage error limit reached.
eSC_ERR_CSC_TIMEOUT	0x2D	CSC communication timeout detected on sensor CAN.
eSC_ERR_CSC_UNDERVOLTAGE	0x01	Cell low voltage error limit reached.
eSC_ERR_CSC_WARNING	0x02	Warning occurred at CSC; for details see sensor CAN.

Error	ErrNr	Description (Short)
eSC_ERR_DATASET	0x3C	CRC error detected in safety controller data set.
eSC_ERR_DATASETCONF	0x3A	Error in data set configuration detected, invalid CRC or version of subcomponents.
eSC_ERR_FLASH_MONITOR	0x23	Error occurred in SC ROM test.
eSC_ERR_FLOW_CTR	0x06	Flow control error between SC and SSV detected.
eSC_ERR_HV_MEAS	0x11	Invalid PMB high voltage measurement values detected.
eSC_ERR_IL_TEST	0x34	Interlock failure detected during start-up test.
eSC_ERR_INTERLOCK_MISSING	0x09	Missing or invalid interlock signal detected.
eSC_ERR_ISO_MEASUREMENT	0x0C	Isolation measurement hardware error detected.
eSC_ERR_ISO_RESISTANCE	0x04	Isolation resistance error limit reached.
eSC_ERR_ISO_SHORT	0x46	Short circuit in the HV lines detected
eSC_ERR_KL30C_TEST	0x36	T30c (KL30c) voltage failure detected during start-up test.
eSC_ERR_LIFEDATA	0x10	Not used
eSC_ERR_MC_WD	0x4A	MC watchdog not triggered or trigger signal out of specification.
eSC_ERR_MSW3_CONTROL	0x43	Not used
eSC_ERR_MSW3_STUCK	0x4D	Not used
eSC_ERR_MSW_AGE	0x27	Main switches reached end of life
eSC_ERR_MSW_AGE_WARN	0x2F	Main switches approaching end of life.
eSC_ERR_MSW_STUCK_TEST	0x40	Main switch stuck test error detected.
eSC_ERR_MSWMINUS_CONTROL	0x41	Error detected in main switch negative control circuit.
eSC_ERR_MSWMINUS_STUCK	0x4B	Main switch negative stuck detected.
eSC_ERR_MSWPLUS_CONTROL	0x42	Error detected in main switch positive control circuit.
eSC_ERR_MSWPLUS_STUCK	0x4C	Main switch positive stuck detected.
eSC_ERR_NVRAM	0x0F	NVRAM error
eSC_ERR_PCB1_TEMPERATURE	0x0D	BMS PCB overtemperature limit reached.
eSC_ERR_PMB_DS_INTEROP	0x44	PMB data set error detected
eSC_ERR_PMB_EMERG_SHORT_SUP	0x22	Short circuit detected between PMB emergency line and supply.
eSC_ERR_PMB_EMERG_TEST	0x35	PMB emergency test failed.
eSC_ERR_PMB_EMERGENCY	0x08	Low PMB emergency line detected.
eSC_ERR_PMB_INTERNAL	0x26	PMB internal error; for details see sensor CAN.
eSC_ERR_PMB_INTEROP	0x32	Invalid PMB application version detected.
eSC_ERR_PMB_LIMITS	0x0E	Defined limit violated at PMB; for details see sensor CAN.

Error	ErrNr	Description (Short)
eSC_ERR_PMB_OC_CHG	0x29	Charge overcurrent limit reached.
eSC_ERR_PMB_OC_DCHG	0x2A	Discharge overcurrent limit reached.
eSC_ERR_PMB_OC_PRECH	0x5D	Overcurrent limit reached during pre-charge
eSC_ERR_PMB_OVT	0x28	Shunt overtemperature limit reached.
eSC_ERR_PMB_OVERVOLTAGE	0x2B	Battery overvoltage error limit reached.
eSC_ERR_PMB_TIMEOUT	0x0A	PMB communication timeout detected on sensor CAN.
eSC_ERR_PMB_WARNING	0x19	Warnings occurred at PMB, for details see sensor CAN.
eSC_ERR_PRE-CHARGE1_DIAG	0x49	Pre-charge control circuit error detected.
eSC_ERR_PRE-CHARGE1_POWER	0x48	Pre-charge resistance power maximum limit reached.
eSC_ERR_PRE-CHARGE2_DIAG	0x50	Not used
eSC_ERR_PRE-CHARGE2_POWER	0x51	Not used
eSC_ERR_RAM_MONITOR	0x17	Error occurred in SC RAM test.
eSC_ERR_REFVOLTAGE	0x1F	Corrupted reference voltage detected on BMS board.
eSC_ERR_RESET_MONITOR	0x15	Illegal SC reset detected.
eSC_ERR_ROM_MONITOR	0x16	ROM monitor
eSC_ERR_RTC	0x47	Unable to read valid time value from RTC.
eSC_ERR_SC_INTERNAL	0x45	Internal SC failure.
eSC_ERR_SPI_COM	0x5A	SSV detected a problem with SPI communication.
eSC_ERR_SSC_ALIVE_TEST	0x31	SSC Alive test failure
eSC_ERR_SSV_CONFIG_HARD	0x1B	SSV data set error detected.
eSC_ERR_SSC_DS_INTEROP	0x59	SSV data set error detected. (Warning state)
eSC_ERR_SSV_EMERGENCY	0x20	SSV detected one or more emergency line events.
eSC_ERR_SSV_INTEROPERABLE	0x0B	Invalid SSV application version detected.
eSC_ERR_SSV_REFVOLTAGE	0x58	SSV detected corrupted reference voltage on BMS board.
eSC_ERR_SSV_STARTUP_TEST	0x3E	SSV start-up test error detected.
eSC_ERR_SSV_SUPPLYVOTLAGE	0x24	Low SSV supply voltage.
eSC_ERR_SSV_TIMEOUT	0x03	Missing communication with SSV detected.
eSC_ERR_SSV_WARNING	0x21	Error in SSV task timing occurred.
eSC_ERR_SSV_WD	0x25	Missing SSV WD trigger signal detected.
eSC_ERR_STACK_MONITOR	0x18	SC stack overflow detected.
eSC_ERR_STARTUP	0x3F	Start-up test error occurred.
eSC_ERR_SUPPLYVOLTAGE	0x1E	Low SC supply voltage detected.
eSC_ERR_TASKSYSTEM	0x53	Internal SC task error occurred.
eSC_ERR_TIMERMANGEMENT	0x52	Internal SC timing error occurred.

Error	ErrNr	Description (Short)
eSC_ERR_TRAC_VOLTAGE	0x12	Traction net overvoltage limit reached.
eSC_ERR_WD	0x13	SSV detected error in trigger line from SC.
eSC_ERR_WD_TEST	0x37	Error occurred during system WD test
eSC_ERR_SSV_ALIVE_TEST	0x31	SSV alive test failed.
eSC_STAT_CLOSE_NOT_POSSIBLE_WEAK	0x60	Cannot perform MSW close request; internal preconditions not met.
eSC_STAT_ISO_GND_WARNING	0x62	ISO ground warning.
eSC_STAT_ISO_RES_WARNING	0x61	ISO resistance warning.

6.2 MC Error Description

EVT_COMPONENT Value: 0x01

Error	ErrNr	Description (Short)
eMC_ERR_DATASET_CRC	0x5	Application data set seems to be corrupt. Redownload the data set and retry.
eMC_ERR_EEPROM	0x6	Error detected during non-volatile data access. Application data could not be read/written.
eMC_ERR_INTER_COM	0x3	Interpack communication warning.
eMC_ERR_INTERNAL	0x2	Internal error.
eMC_ERR_MASTER_SLAVE_CONFIG	0x4	Invalid network configuration detected. Please configure the hardware pins of the BMS correctly. Configuration is described in the specific technical documentation of the Kreisel Battery Pack
eMC_ERR_MSW_CLOSE_NOT_POSSIBLE	0x0	Main switch close request will not be handled because of active MC warnings.
eMC_ERR_TIMEOUT_MSG_REQ	0x1	EXT_FuncRequest message timeout.

6.3 SC Error Description

EVT_COMPONENT Value: 0x03

Error	ErrNr	Description (Short)
ePMB_ERR_BROKEN_WIRE	0x37	Wire break detected.
ePMB_ERR_CAN_ERROR_ID	0x2D	Wrong CAN ID received.
ePMB_ERR_CPU_MONITOR	0x35	CPU monitoring error.
ePMB_ERR_CURRENT_CHARGE_MAX	0x21	Overcurrent charge limit occurred.
ePMB_ERR_CURRENT_DISCHARGE_MAX	0x20	Overcurrent discharge limit occurred.
ePMB_Error_I_OCD_CHARGE_OOR	0x2B	Overcurrent detection charge value out of range.
ePMB_Error_I_OCD_DISCHARGE_OOR	0x2A	Overcurrent detection discharge value out of range.
ePMB_ERR_EEPROM_READ_WRITE	0x3A	Unable to read/write EEPROM parameter.

Error	ErrNr	Description (Short)
ePMB_ERR_FLASH_CRC_A_DATA_RECORD	0x19	CRC of data record (data set) A failed.
ePMB_ERR_FLASH_CRC_B_DATA_RECORD	0x1A	CRC of data record (data set) B failed.
ePMB_ERR_FLASH_CRC_CALIB_PARA	0x16	CRC of calibration parameter failed.
ePMB_ERR_FLASH_CRC_CAN_PARA	0x0D	CRC of CAN parameter failed.
ePMB_ERR_FLASH_CRC_IHM_PARA	0x13	CRC of IHM data failed.
ePMB_ERR_FLASH_CRC_PROD_DATA	0x10	CRC of production data failed.
ePMB_ERR_FLASH_READ_CALIB_PARA	0x17	Unable to read calibration parameter.
ePMB_ERR_FLASH_READ_CAN_PARA	0x0E	Unable to read CAN parameter.
ePMB_ERR_FLASH_READ_IHM_PARA	0x14	Unable to read IHM ASIC parameter.
ePMB_ERR_FLASH_READ_PROD_DATA	0x11	Unable to read production data.
ePMB_ERR_FLASH_WRITE_CALIB_PARA	0x18	Unable to write calibration parameter.
ePMB_ERR_FLASH_WRITE_CAN_PARA	0x0F	Unable to write CAN parameter.
ePMB_ERR_FLASH_WRITE_IHM_PARA	0x15	Unable to write IHM ASIC parameter.
ePMB_ERR_FLASH_WRITE_PROD_DATA	0x12	Unable to write production data.
ePMB_ERR_IHM_CAR_CHECK	0x05	IHM ASIC register CAR check failed
ePMB_ERR_IHM_INIT_CAR	0x09	Write and read IHM ASIC register CAR failed
ePMB_ERR_IHM_INIT_CRA	0x0C	Write and read IHM ASIC register CRA failed
ePMB_ERR_IHM_INIT_CRG	0x0B	Write and read IHM ASIC register CRG failed
ePMB_ERR_IHM_INIT_OPM	0x08	Write and read IHM ASIC register OPM failed
ePMB_ERR_IHM_INIT_TRR	0x0A	Write and read IHM ASIC register TRR failed
ePMB_ERR_IHM_TRR_CHECK	0x06	IHM ASIC register TRR check failed
ePMB_ERR_IHM_ZZR_CHECK	0x07	IHM ASIC register ZZR check failed
ePMB_ERR_INTERRUPT_MONITOR	0x30	Interrupt monitoring error.
ePMB_ERR_MEAS_CURRENT_OUFLOW	0x1B	Current measurement overflow.
ePMB_ERR_MEAS_TEMPINT_OUFLOW	0x1F	Internal temperature measurement overflow
ePMB_ERR_MEAS_VOLTAGE_BAT_OUFLOW	0x1C	Battery voltage measurement overflow.
ePMB_ERR_MEAS_VOLTAGE_TRAC_N_OUFLOW	0x1E	Traction- voltage measurement overflow.
PMB_Error_MEAS_U_TRAC_P_OUFLOW	0x1D	Traction+ voltage measurement overflow
PMB_Error_MSW_AGE_CRC	0x3C	CRC error at main switch aging data.
PMB_Error_MSW_AGE_ERR	0x39	Main switch cycle counter error limit occured.
PMB_Error_MSW_AGE_WRN	0x38	Main switch cycle counter warning limit occured.
PMB_Error_OCD	0x01	Overcurrent detected during charge/discharge process.
PMB_Error_RAM_MONITOR	0x33	Error at RAM monitoring.
PMB_Error_RESET_MONITOR	0x31	Error at RESET monitoring.
PMB_Error_ROM_MONITOR	0x34	Error at ROM monitoring.

Error	ErrNr	Description (Short)
PMB_Error_SEND_CAN	0x2E	CAN messages can't be sent.
PMB_Error_STACK_MONITOR	0x32	Error at Stack monitoring.
PMB_Error_SYS_GENERAL	0x04	A system error occurs on the μ C
PMB_Error_TASK_TIME	0x2C	Task time overruns its time.
PMB_Error_TEMPERTURE_MAX	0x28	Over temperature limit reached.
PMB_Error_TEMPERTURE_MIN	0x29	not used
PMB_Error_U_REF	0x36	Reference voltage out of limit.
PMB_Error_UBAT_COMPARE_IHM_UC	0x2F	Battery voltage of IHM-ASIC and μ C differs
PMB_Error_VOLTAGE_BAT_MAX	0x22	Limit for batterie voltage reached.
PMB_Error_VOLTAGE_BAT_MIN	0x23	not used
PMB_Error_VOLTAGE_TRAC_N_MAX	0x26	Limit for traction- voltage reached.
PMB_Error_VOLTAGE_TRAC_N_MIN	0x27	not used
PMB_Error_VOLTAGE_TRAC_P_MAX	0x24	Limit for traction+ voltage reached.
PMB_Error_VOLTAGE_TRAC_P_MIN	0x25	not used
PMB_Error_WRONG_ARTICLE_NUMBER	0x02	Wrong article number.
PMB_Error_WRONG_DATARECORD	0x03	CRC of the datarecord(dataset) A or/and B are wrong.
ePMB_ERR_FLAG	0x0	One of the errors is active

6.4 SC Error Description

EVT_COMPONENT Value: 0x04 to 0x13

Error	ErrNr	Description (Short)
eCS C_ERR_001_EMERGENCY_LINE	0x1	Emergency line set.
eCSC_ERR_002_ARTICLE_NUMBER	0x2	Invalid article number. /Unable to read article number.
eCSC_ERR_003_LIM_OVERVOLT	0x3	Cell overvoltage limit reached.
eCSC_ERR_004_LIM_UNDERVOLT	0x4	Cell undervoltage limit reached.
eCSC_ERR_005_LIM_OVERTEMP	0x5	Cell overtemperature limit reached
eCSC_ERR_006_LIM_UNDERTEMP	0x6	Minimum temperature limit reached (-60°C)
eCSC_ERR_007_LIM_OVERTEMP_PCB	0x7	PCB overtemperature limit reached
eCSC_ERR_008_LIM_CONTACT_RES	0x8	Cell contact resistance error detected.
eCSC_ERR_009_LIM_POWER_SUPPLY	0x9	Power supply error detected.
eCSC_ERR_010_LIM_INT_DIE_TEMP	0xA	Temperature limit 125°C reached
eCSC_ERR_012_MEAS_INIT	0xC	Measurement initialization error detected.
eCSC_ERR_014_MEAS_READ_DATA	0xE	Unable to read measured data. Unable to read data from measurement boards.
eCSC_ERR_016_MEAS_WRITE_CMD	0x10	Measurement board write command failed
eCSC_ERR_017_MEAS_SET_CONFIG	0x11	Unable to set measurement board configuration.

Error	ErrNr	Description (Short)
eCSC_ERR_018_MEAS_SEQ	0x12	Measurement sequence failed.
eCSC_ERR_019_MEAS_LTC_TESTS	0x13	LTC test failed
eCSC_ERR_024_BAL_WRITE_CFRG	0x18	Balancing write configuration error.
eCSC_ERR_025_BAL_COMPARE_CFRG	0x19	Balancing compare configuration error.
eCSC_ERR_026_BAL_THRESHOLD	0x1A	Balancing threshold voltage error. Balancing voltage reached limit.
eCSC_ERR_027_BAL_CONTACT_RES	0x1B	Balancing contact resistance error. Unable to complete balancing correctly due to bad cell contact.
eCSC_ERR_029_SYS_GENERAL	0x1D	System general error
eCSC_ERR_030_SYS_SWITCH_DEF	0x1E	Program internal error on switching cell voltage measurement.
eCSC_ERR_031_SYS_DIFF_ZERO	0x1F	Configured number of voltages 0.
eCSC_ERR_032_SYS_TASK_TIME	0x20	Task time error.
eCSC_ERR_033_SYS_DATASET_WARN	0x21	One of the two data set sections has an invalid CRC.
eCSC_ERR_034_SYS_DATASET	0x22	Both data set sections have an invalid CRC.
eCSC_ERR_037_FLASH_CRC_CAN	0x25	Flash CRC incorrect.
eCSC_ERR_038_FLASH_READ_CAN	0x26	Unable to read Flash data.
eCSC_ERR_039_FLASH_CRC_PROD	0x27	Production data Flash CRC incorrect.

7 Notes



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